

Teaching Scheme of Work

Module title:	Research Methods and Geographical Information Systems
Courses accessing the module:	BSc (Hons) Environmental Management and Sustainability, FdSc Countryside Management
Module code:	C5025HF
Credit (Learning Hours):	

	Name	E-mail	Office Locaion(s)
Module Leader:	Joseph Mhango		
Other Tutors:			

Assessment:	
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Module Learning Outcomes

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The module resource list for this module can be found here:	(please instert link to HARLO resource list here)
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Week	Session Type	EVENT DURATION	NUMBER OF SESSIONS	THEME/TOPIC	Student Preparation for Class	Independent Class Follow Up	Recommened Assesement Preparation
1	Lecture	2	1	Introduction to the module and Information literacy			
	Tutorial	1	1	Effective searching and literature review			
	Practical	0	0	0			
2	Lecture	2	1	Research strategies and design for social sciences			
	Tutorial	2	1	Questionnaire design			
	Practical	0	0	0			

3	Lecture	2	1	Research and the scientific method, Probability distributions			
	Tutorial	2	1	Descriptive statistics: Measures of central tendency, dispersion, confidence intervals (R-based tutorial)			
	Practical	0	0	0			
4	Lecture	2	1	Parametric and non-parametric hypothesis testing			
	Tutorial	2	1	Test statistics and their applications: T-tests, X2 tests, F tests (Excel-based tutorial)			
	Practical	1	1	Assignment support: Qand A			
5 (optional independent study level 4)	Lecture	2	1	Introductory R			
	Tutorial	2	1	R tutorial: T-tests, X2 tests, F tests (R-based tutorial)			

	Practical	0	0	0			
6 (reading week level 5 and 6)	Lecture	0	0	Independent Study			
	Tutorial	0	0	0			
	Practical	0	0	0			
7	Lecture	2	1	0			
	Tutorial	2	1	ANOVA tutorials (R-based tutorial)			
	Practical	0	0	#REF!			
	Lecture	2	1	Non-parametric statistics			

8	Tutorial	2	1	Non-parametric statistics tutorial			
	Practical	0	0	0			
9	Lecture	2	1	Spatial Data & Mapping (vector data, raster data, attributes, visualisation)			
	Tutorial	2	1	Working with attribute tables - filtering, joins, intersections,			assignment submission
	Practical	0	0	0			
10 (reading week level 6)	Lecture	2	1	Spatial Data & Mapping: Interpolation and map creation			
	Tutorial	2	1	Interpolation, creation of maps			
	Practical	0	0	0			

11	Lecture	2	1	Sampling theory: Design vs model-based sampling			
	Tutorial	2	1	Implementing sampling methods on spatial data in R (convenient, random, stratified, cluster sampling e.t.c			
	Practical	0	0	0			
VACATION							
12	Lecture	2	1	Explaining variance in data: Regression and correlation			
	Tutorial	2	1	Regression and correlation tutorials (R-based tutorial)			
	Practical	0	0	0			
	Lecture	2	1	Communicating research			

16	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
17	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
18 (reading week level 6)	Lecture	0	0	0			
	Tutorial	0	0	0			

	Practical	0	0	0			
19	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
20	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
	Lecture	0	0	0			

21	Tutorial	0	0	0			
	Practical	0	0	0			
22	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
VACATION							
23	Lecture	0	0	0			
	Tutorial	0	0	0			

	Practical	0	0	0			
24	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
25 (reading week level 5 and 6)	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
	Lecture	0	0	0			

26	Tutorial	0	0	0			
	Practical	0	0	0			
27	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			
28 (reading week level 6)	Lecture	0	0	0			
	Tutorial	0	0	0			
	Practical	0	0	0			

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